

[DC] Mixed Reality for Psychological Resilience: A Conceptual Framework for Rescue Training in combination with ACT

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ABSTRACT

The digital transformation offers novel training opportunities for emergency and rescue teams. Traditional simulations in volunteer services face limitations, often lacking either visual realism or essential physical feedback. This PhD project addresses this gap by proposing a Mixed Reality training model. By combining physical training dummies with responsive virtual patients, the system allows for the simultaneous practice of procedural skills and mental resilience under realistic psychological stress. The goal is to integrate technical proficiency with enhanced mental well-being using Acceptance and Commitment Therapy principles.

Index Terms: Extended reality, digital therapeutics, acceptance and commitment therapy, gamification.

1 INTRODUCTION

The world is in a constant state of change, just as we humans are. We are constantly exposed to significant shifts in technology, society, and the economy. While these changes present opportunities for further development, they can also sometimes feel like a threat. In this context, innovative technologies like Extended Reality (XR) have proven to be an effective tool over the last few decades in various fields, including industry and education. XR can offer the opportunity to create an immersive and interactive 3D experience by simulating environments. This makes complex cases tangible and directly experienceable. The technology moves beyond a traditional simulation environment, enabling real-time interaction and the collection of body-specific data. Sensors and cameras can capture real-time metrics such as movement data, gaze patterns, and facial expressions. This fundamentally transforms the user from a passive observer into an active participant. Against this backdrop, the goal of this dissertation is to develop a conceptual framework that uses established therapeutic methods like Cognitive Behavioral Therapy (CBT), Acceptance and Commitment Therapy (ACT), and Schema Therapy, combining them with XR into a modular approach to psychological resilience training. The goal is to practice both procedural skills and mental resilience at the same time, while managing realistic psychological stress.

2 RELATED WORK

Digital transformation is not only evident in technology but also in healthcare. Lifestyle and health apps on smartphones have become increasingly important in recent years and are an integral part of everyday life for many people. This is also supported by a survey of German smartphone users aged 16 and over, which found that mental health apps are among the top five most frequently used categories of apps [8, 1]. Parallel to the digital transformation, there is also a new era in modern psychotherapy. Process-Based

Therapy (PBT) is increasingly challenging the traditional medical disease model, which views psychological illnesses as clearly defined entities to be treated with standardized treatment protocols, primarily focusing on symptom reduction [7, 5]. Instead, the focus in the future will be on core psychological processes that address the maintenance or prevention of mental health. This should be supported by a set that is individually created using the modular methodology toolbox of PBT [2]. Thus, the PBT approach offers a flexible, transdiagnostic perspective that responds to the individual needs of those affected [5]. In this context, the methods of ACT will be used. ACT is categorized as part of the fourth wave of behavioral therapy, and its broad effectiveness in treating psychological disorders is well documented [3]. The main goal of ACT is to foster psychological flexibility in the face of seemingly unchangeable adversities (acceptance) to stimulate self-compassion and to allow for value-driven action even under hardship (commitment). The combination of modern psychological approaches and new technologies has given way to a new form of Digital Therapeutics (DTx) [6]. Here, immersive technologies like Mixed Reality (MR) offer new potential because, unlike Virtual Reality (VR) or Augmented Reality (AR), they bridge the gap by allowing digital and physical objects to be seamlessly integrated and interacted with. As a result, MR offers more than traditional simulation, which quickly becomes limited in terms of safety, costs, and repeatability [4]. However, the introduction of MR-supported therapies also brings challenges. These include a high implementation effort and technical complexity. Furthermore, problems related to the user, such as cognitive overload and the novelty effect, as well as physiological reactions like cybersickness (which can manifest as dizziness and nausea), can limit acceptance and applicability.

3 METHODOLOGY

This research focuses on developing a training approach that specifically strengthens the cognitive load capacity and psychological resilience of emergency staff. For the rest of this paper, this approach will be referred to as Mixed Reality - Advanced Rescue Training (MR-A.R.T.).

Particularly in volunteer services, like the fire department, medical aid, or disaster relief, traditional training methods often run into an invisible boundary between physical reality and virtual simulation. The problem is usually that the training is either haptically realistic, like on a training dummy, but visually 'empty' or visually well-done but not haptically realistic, which is bad for learning manual skills. In real-world situations, however, these people must work precisely under massive visual and auditory stress and still have to think clearly. The goal of MR-A.R.T. is to combine technical-professional training, ACT-based resilience and skill training, and MR in such a way that trainees learn to perform their tasks safely - even under high psychological stress, while take care of their mental health and do not take any real risks. The following research questions (RQs) and hypotheses (H) are derived based on this approach:

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RQ 1: How can modern technologies like XR, in combination with ACT, be used to effectively raise awareness among emergency personnel and provide them access to psychological training? H_0 : There is no significant difference in effectiveness or accessibility between XR-based and traditional training methods. H_1 : The immersive nature of MR combined with ACT leads to significantly higher engagement compared to traditional methods. By reducing the cognitive load required to imagine scenarios, participants can focus more effectively on the exercises, thereby increasing their resilience against related stressors and potential trauma.

RQ 2: How does gamification influence the therapeutic process, especially regarding user motivation and involvement? Both positive and potential negative effects of XR should be examined. H_0 : Gamification elements in XR have no significant impact on user motivation or engagement. H_1 : Gamification significantly increases motivation and adherence, but if the game mechanics are too complex, they can lead to cognitive overload, distracting trainees from the core content and lowering the immediate learning outcome.

The core idea of this framework is to use MR combined with inside-out trackers to visually transform a static training dummy into a dynamic patient. This allows adjusting the patient's appearance (age, gender, injury pattern, and so on) as well as the medical progression in real-time. You can even talk to the dummy, and it can respond, which can help you imagine what the situation could be like. This training concept has an advantage compared to traditional methods, instead of getting shocked by sudden stress in a real emergency, MR can expose trainees to controlled, gradual stress. The brain gets the possibility to adapt to the simulated chaos, and if a real situation happens, the brain can process it as a pattern he already knows instead of seeing it as a new threat. This may lower the risk of acute stress reactions. For a further enhancement, I plan to use eye-tracking. This technology allows for the generation of a heatmap of the trainee's gaze pattern, providing objective data on where they were truly focusing throughout the scenario. This helps confirm whether they were concentrating on the real task they had to do or being distracted. As an additional option to be explored is using a commercial heart rate monitor from sports to measure various signals from the body that show how stressed the trainees are during the different parts of the exercises. Furthermore, if you successfully dealt with a 'real' crisis, the user will become more confident. They will feel less worried about failing because they will remember that they have already done it successfully. Finally MR creates a 'safe space' for making mistakes. Trainees can experience how it feels to make mistakes like 'losing' a patient without the real-life consequences, which lets them process these feelings safely, helps practicing psychological skills and reduces the overwhelming anxiety of real-life responsibility. Table 1 shows how rescue scenarios can be used in the six main processes of ACT to help emergency personnel handle difficult situations.

4 CONCLUSION

The present dissertation approach proposes an original conceptual framework, the aim of which was to identify a method of reducing the discrepancy between physical skills training and psychological stress resilience in volunteer emergency services. By integrating ACT into an MR-based environment, the goal is to develop an approach that addresses the limitations of traditional simulation methods. While conventional procedures often neglect either haptic reality or visual immersion, the overlay of physical training dummies with dynamic, virtual content described here enables holistic training. The objective is to prioritize the human aspect of emergency response and use technology to train not only processes but also mental resilience.

Table 1: ACT Processes and their usage for rescue scenarios

ACT Technique	Psychological Challenge	Goal in MR-A.R.T. Training
Cognitive Defusion	Continuous struggle with catastrophic or self-critical thoughts	Learning to gain distance from thought patterns that block action; recognizing that thoughts are not facts, that they do not need suppression and that they do not prevent actions
Acceptance	Avoidance of aversive states (fear, dizziness, rapid heartbeat) that lead to hesitation	Being prepared to tolerate physical discomfort and keep going
Present Moment	Distraction by past mistakes or worries about future risks	Training the focus of attention on the immediate task in the 'here and now'
Self as Context	Excessive identification with failures, e.g., 'I am a poor rescuer.'	Getting an objective perspective without linking it to the success or failure of the exercise
Values and Committed Action	Deviation from the protocol due to exhaustion or fear	Continuing and focusing on an efficient and safe rescue approach within all obstacles

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